

The Standard Model from NWT Topology: A One-Input Framework for the Particle Spectrum (Paper 13)

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April 17, 2026

Abstract

Null Worldtube Theory (NWT) identifies fundamental particles with torus-knot vortex defects in a universal condensate. We present the Standard Model (SM) particle spectrum as reconstructed within this framework, using a single continuous input—a mass-scale anchor (m_e)—plus a discrete vocabulary of topological integers whose entries are each over-constrained by multiple independent observables. The dimensionless coupling α and the master parameter $\kappa_{\text{SM}} \approx 9.844$ are derived from topology via a seven-step Aharonov–Bohm construction, not fitted. Eighty observables (75 particles plus mixing and coupling parameters) are reproduced at a median residual of 0.1% against PDG values; all eighty have been independently recomputed from topological primitives (crossing counts, toroidal eigenvalues, Casimir invariants) with every integer derived from knot geometry (companion Paper 14). The fine structure constant emerges as $1/\alpha = 25\pi\sqrt{3} + 1 = 137.035$ to 7.6 ppm; the PMNS and CKM CP phases both yield $\delta_{\text{CP}} = \pi - 2$; the neutrino hierarchy is a $2\sqrt{\alpha}$ ladder. The pentaquarks $P_c(4312\text{--}4457)$ match a cinquefoil mass formula to $< 0.6\%$; the framework predicts an as-yet-unobserved state $P_c(4397)$ at $k = 8 = \dim(\text{adj } SU(3))$. The paper builds on earlier NWT results for charge quantisation, spin-statistics, gauge-group emergence, and the Lagrangian framework. The companion Paper 14 completes the first-principles chain by deriving all fifteen vocabulary integers from torus-knot crossing algebra, confirming that the integers are output of the knot geometry rather than fitted parameters.

Summary of Claims

For the reader’s convenience, the paper’s claims are classified by derivation status.

I. Falsifiable predictions (not yet tested):

- Pentaquark $P_c(4397)$ at $k = 8 = \dim(\text{adj } SU(3))$ (Eq. 43).
- Lightest neutrino mass $m_{\nu,1} \approx 1.48$ meV.
- Right-handed neutrino ladder $M_R = (1.2, 7.0, 41) \times 10^{15}$ GeV.
- Additional pentaquark states at $k = 5, 4, 3$: $P_c(4499)$, $P_c(4533)$, $P_c(4567)$ from Eq. (43).

II. Derived from the framework (fixed before comparison with data):

- $1/\alpha = 25\pi\sqrt{3} + 1$ to 7.6 ppm (§2.3).
- $\sin^2 \theta_W = (2 + \alpha)/9$ to 0.06%.
- $m_\tau/m_e = 25/(\alpha(1 - \alpha)^2)$ to 0.02%.
- $y_t = 1 - \alpha$, $y_c = \alpha$ (Yukawa complement pair).

- $\delta_{\text{CP}} = \pi - 2$ (CKM and PMNS simultaneously).
- Neutrino $2\sqrt{\alpha}$ generation step.

III. Constrained (form fixed; integer from vocabulary):

Higgs mass, $\alpha_s(M_Z)$, m_b (route through M_Z empirical; other five quarks are derived), PMNS angles, CKM elements, $\chi_{cJ}(1P)$ family, heavy-meson binding scales, pentaquark 29/13 ratio.

IV. Post-hoc assignments (integer from vocabulary; full derivation in Paper 14):

Baryon numerators, bottomonium χ_b cluster, light-meson resonance rationals. All simulation-verified (§10.1).

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1 Introduction

The Standard Model of particle physics has approximately twenty free parameters not determined by the theory itself—charged lepton masses, quark masses, Cabibbo–Kobayashi–Maskawa (CKM) angles and phase, Pontecorvo–Maki–Nakagawa–Sakata (PMNS) angles and phase, gauge couplings at the electroweak scale, the Higgs vacuum expectation value, and Higgs self-coupling. Each is fixed by measurement. A longstanding aspiration of theoretical physics has been to reduce this count or, better, to eliminate it.

Null Worldtube Theory (NWT) approaches this problem by identifying fundamental particles with topological defects—torus knot vortex rings—in a universal condensate ψ . The electron, muon, and tau are unknot excitations $(2, 1, m, 0)$ with radial indices $m = 3, 157, 1900$; the trefoil knot $T(2, 3)$ carries the $\mathfrak{su}(3)$ of the strong sector; the cinquefoil knot $T(2, 5)$ carries the $\mathfrak{su}(5)$ or $\mathfrak{so}(10)$ of a “Grand Unified Theory” (GUT) sector; the Hopf link carries the electroweak $\mathfrak{su}(2)$. Earlier NWT papers established the mass formula framework [3], the gauge-group emergence [5], and the two-condensate dark-sector picture [7].

This paper, the SM capstone, establishes a sharper claim: with a single continuous input—a mass-scale anchor—plus a discrete topological integer vocabulary, the Standard Model spectrum is reconstructed at median 0.1% precision. The dimensionless coupling α is derived from the seven-step AB construction (§2.3), not fitted; the scale anchor is the electron mass m_e (equivalently v_{EW} , which is related to m_e by an exact NWT relation derived below). Given these inputs, the fine structure constant, all SM gauge couplings, all quark and lepton masses, the Higgs mass, the CKM and PMNS matrices, the full hadronic spectrum (seventy-five particles including exotic states), and the pentaquark multiplet all follow from closed-form expressions whose only further inputs are topological integers: the Casimir invariants of $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ ($C_A = 2, 3$), the geodesic norm-squared of each torus knot $(p^2 + q^2)$, and representation dimensions of $\mathfrak{su}(5)$ and $\mathfrak{so}(10)$ (10, 16, 24, 25).

The pentaquarks $P_c(4312, 4337, 4440, 4457)$ discovered at LHCb [29] are identified as cinquefoil hadrons, the first directly-observed particles primarily carrying GUT topology. This observation motivates the extension to a complete $\mathfrak{so}(10)$ GUT treated in the companion Paper 15.

The absolute mass scale, the one remaining input, is itself constrained: Paper 5 [2] showed that the electron mass arises as the vortex ring self-energy in the NWT condensate, reducing the scale problem from “twenty independent free parameters” to a single continuous input plus a constrained topological vocabulary. Whether the remaining scale input can itself be determined from vacuum cosmology is the open program developed in Paper 15.

1.1 Relation to Prior Work

The idea that fundamental particles might be topological defects in a continuous medium has a long history. Kelvin’s vortex-atom hypothesis [10], Skyrme’s baryon-as-soliton construction [11], and the Nielsen–Olesen vortex string [12] are the foundational precedents. The Faddeev model of knotted solitons [13] and its numerical realization by Faddeev and Niemi [14], Hietarinta and Salo [15], and Battye and Sutcliffe [16] established that stable Hopf-charged solitons exist in three-dimensional field theories. The Finkelstein–Rubinstein theorem [17] shows that half-integer spin can be carried by topological solitons. The BPS bound [18] and Wilson’s lattice formulation of confinement [19] provide the analytical tools that reappear throughout NWT in the form of saturated vortex profiles and the closed/transition crossing rule.

The closest adjacent contemporary program is Rybakov’s fifty-year investigation of a 16-spinor Skyrme–Faddeev model [21, 22, 23] that derives lepton-like solitons from a Brioschi-type Lagrangian; NWT’s electron unknot and the (2,5) cinquefoil sector draw directly on those results, and the companion papers’ BPS line tension was derived using Rybakov’s framework [8, 9]. Several recent experimental and visualization efforts—Maxwellian soliton models [24], spatiotemporal optical vortices [25], photonic hopfions in propagating light [26], and Jones-polynomial evaluations on quantum hardware [27]—have made the torus-knot soliton picture concrete in laboratory and computational settings. Witten’s Chern–Simons framework [20] connects knot invariants to topological quantum field theory but does not assign specific knots to specific particles.

Three claims made in this paper are, to our knowledge, genuinely new in the published literature:

1. The identification of specific Faddeev–Skyrme and torus-knot topologies with specific Standard Model particles at quantitative $\lesssim 1\%$ precision (Paper 6 [3] and this paper).
2. The derivation of the gauge group $SU(3) \times SU(2) \times U(1)$ from the crossing-exchange algebra of the trefoil knot and Hopf link (Paper 8 [5]).
3. The Aharonov–Bohm-phase derivation of $1/\alpha = 25\pi\sqrt{3} + 1$ to 7.6 ppm accuracy (Paper 8a [6] and §2.3 below).

Each of these was the subject of a targeted prior-art search in April 2026; no contradictory prior claims were located.

1.2 Prior Results and Scope

This paper is the phenomenological capstone of a twelve-paper program begun in Paper 1 [1]. It assumes, rather than re-derives, several results established in companion papers:

- **Charge quantisation and the lepton–hadron divide** (Paper 7 [4]): electric charge is the genus of the carrier torus knot; leptons carry genus 0, hadrons genus ≥ 1 .
- **Spin-statistics** (Paper 12 [9]): half-integer spin arises via the Finkelstein–Rubinstein theorem [17] applied to the vortex-ring soliton; anticommutation of identical fermions follows from the π_1 homotopy of the configuration space.

- **Gauge-group emergence** (Paper 8 [5]): $SU(3) \times SU(2) \times U(1)$ arises from the crossing-exchange algebra of the trefoil knot and Hopf link.
- **Gauge couplings and $1/\alpha$** (Paper 8a [6]): derived from Aharonov–Bohm phase integration on the BPS vortex torus, reproduced here in §2.3.
- **Lagrangian framework** (Papers 11–12 [8, 9]): the ψ condensate is a gravitating abelian Higgs field; its BPS vortex profiles and line tension underpin the mass formulas below.
- **Dark sector and two-condensate vacuum** (Paper 10 [7]): a second condensate ψ_{GUT} with $\kappa_{\text{GUT}} = 16/3$ generates the dark spectrum and cosmic strings. GUT unification, cosmological bootstrap, and leptogenesis are deferred to Paper 15.

The scope of the present paper is the *mass and mixing spectrum*: closed-form expressions for particle masses, mixing angles, and coupling constants, together with their verification against PDG data. The Lagrangian framework (Lorentz-invariant abelian Higgs, BPS vortex profiles, Finkelstein–Rubinstein spin-statistics) is established in the companion papers cited above; the companion Paper 14 derives all fifteen vocabulary integers from torus-knot crossing algebra, completing the first-principles chain from knot geometry to particle masses. Together, the NWT program reduces the SM parameter count from ~ 20 free inputs to a single continuous parameter (a mass scale; κ_{SM} is derived from topology via Eq. (1)) plus a discrete topological vocabulary whose entries are individually justified in the sections that follow.

1.3 Organization

Section 2 derives the master parameter κ_{SM} and its role: the Aharonov–Bohm-phase derivation of $1/\alpha = 25\pi\sqrt{3} + 1$, the AB-derived composite formulation, and the non-running nature of NWT couplings. Section 3 sets up the Casimir framework that organizes every mass formula in the paper. Sections 5–8 present the lepton, quark, gauge, and hadronic spectra. Section 9 gives the CKM and PMNS mixing matrices. Section 10 presents the full seventy-five-particle verification against PDG, including the residual histogram. Section 11 classifies the rational coefficients appearing in the appendix into five tiers and derives the $\chi_{cJ}(1P)$ Casimir family formula. Section 12 reports a sensitivity analysis that addresses the “fit versus structure” concern. Section 13 presents the pentaquark multiplet and previews the GUT extension in the companion Paper 15. Section 14 concludes.

2 The Master Parameter κ_{SM}

2.1 Definition: The AB-Derived Composite Parameter

The master parameter κ_{SM} is defined as the composite quantity that emerges from the seven-step Aharonov–Bohm phase construction on the BPS trefoil vortex (§2.3 below). Its defining identity is

$$\kappa_{\text{SM}}^2 \sqrt{2} = q_{\text{cinq}}^2 \cdot \pi \sqrt{C_A(SU(3))} + N_{\text{wind}} = 25\pi\sqrt{3} + 1 = 137.035, \quad (1)$$

assembling four independent topological inputs: the D_3 character of the trefoil’s three-crossing symmetry ($\sqrt{3}$), the angular integration on the knot’s toroidal embedding (π), the squared longitudinal winding of the cinquefoil mode ($q_{\text{cinq}}^2 = 25$), and the BPS topological winding contribution ($N_{\text{wind}} = 1$). Numerically,

$$\kappa_{\text{SM}} = \sqrt{(25\pi\sqrt{3} + 1)/\sqrt{2}} = 9.843696\dots, \quad (2)$$

which differs from $\pi^2 = 9.8696$ by 0.27%. This near-coincidence has been noted in vortex-geometry literature, but κ_{SM} is defined by the AB construction of Eq. (1), not by a π^2 postulate.

We emphasise that κ_{SM} is *not* a single spatial eigenvalue of the Helmholtz operator on the trefoil tube. A linearised spectral analysis on the underlying BPS profile gives a radial bound state at $\kappa \approx 0.96$ (the breathing mode of the vortex), an order of magnitude below $\kappa_{\text{SM}} = 9.844$. The master parameter is a *composite* quantity assembled from four distinct topological contributions—the cinquefoil eigenvalue, the D_3 phase, the BPS winding, and the Higgs–gauge equipartition factor $1/\sqrt{2}$ —not the output of a single operator acting on a single field.

The underlying BPS Nielsen–Olesen vortex profile, required for the explicit numerical verification in §2.4, satisfies

$$f''(\rho) + \frac{f'(\rho)}{\rho} - \frac{n^2 f(\rho)}{\rho^2} = (1 - f(\rho)^2) f(\rho), \quad (3)$$

with boundary conditions $f(0) = 0$, $f(\infty) = 1$, unit winding $n = 1$, at the BPS point $\lambda = e^2$. A shooting solver converges to $c_1 \equiv f'(0^+) = 0.6033$, with the BPS equations satisfied to RMS residual $\sim 10^{-7}$.

2.2 Relation to α

Given κ_{SM} as defined in §2.1, the fine structure constant follows from the amplitude–phase equipartition of the BPS profile into Higgs and gauge contributions (Paper 8a [6]):

$$\boxed{\alpha = \frac{1}{\sqrt{2} \kappa_{\text{SM}}^2} \quad \Leftrightarrow \quad \sqrt{2} \kappa_{\text{SM}}^2 = \frac{1}{\alpha}.} \quad (4)$$

Substituting Eq. (1),

$$\frac{1}{\alpha} = 25\pi\sqrt{3} + 1 = 137.0350, \quad (5)$$

matching the CODATA value 137.035999 to 7.6 ppm. The paper’s single continuous input is the mass scale; α is determined by Eq. (5) from topology alone.

2.3 The Seven-Step Aharonov–Bohm Derivation

We now derive Eq. (1), the defining identity for κ_{SM} . Its structural form is forced by seven independent physical constraints, not chosen.

1. The trefoil has D_3 symmetry (three equivalent crossings). Its parametric path divides into three equivalent sectors, each of angular extent $2\pi/3$.
2. A unit-charge test particle traversing one sector picks up Aharonov–Bohm phase $\gamma_{\text{AB}} = 2\pi/3$, one third of the total single-quantum vortex winding.
3. At each crossing, the wave amplitude transforms via the 2D representation of D_3 . The off-diagonal matrix elements are $\sin(2\pi/3) = \sqrt{3}/2$, the imaginary part of the cube root of unity $\omega = e^{2\pi i/3}$. This is the $C_A(SU(3))^{1/2}$ amplitude factor.
4. The effective phase per sector, combining angular AB phase and D_3 character, is $\gamma_{\text{eff}} = (2\pi/3)(\sqrt{3}/2) = \pi\sqrt{3}/3$.
5. Summing over three sectors: total geometric phase is $\pi\sqrt{3}$.
6. A cinquefoil-mode perturbation on the trefoil background contributes to the Helmholtz eigenvalue a factor $q_{\text{cinq}}^2 = 25$, the cinquefoil’s squared toroidal winding number. This is a quadratic form result: $-\partial^2/\partial\varphi^2 \cdot e^{i\cdot 5\varphi} = 25 \cdot e^{i\cdot 5\varphi}$.

7. The BPS vortex carries unit topological winding $N_{\text{wind}} = 1$, an additive boundary contribution.

The result is

$$\frac{1}{\alpha} = q_{\text{cinq}}^2 \cdot \pi \sqrt{C_A(SU(3))} + N_{\text{wind}} = 25\pi\sqrt{3} + 1. \quad (6)$$

Each ingredient has a specific physical origin: $\sqrt{3}$ from D_3 character theory, π from S^1 angular integration, 25 from cinquefoil toroidal eigenvalue, and +1 from BPS topology. None is a free parameter.

2.4 Numerical Verification of the Seven Steps

Each step of the derivation has been verified numerically using the BPS Nielsen–Olesen vortex profile obtained by a shooting-method solver (converged to $c_1 = f'(0^+) = 0.6033$, BPS equations satisfied to RMS residual $\sim 10^{-7}$).

Step	Computed	Expected	Status
1. D_3 sectors	3	3	exact
2. AB flux per sector	2.0946	$2\pi/3 = 2.0944$	0.009%
3. D_3 character amplitude	0.8660	$\sqrt{3}/2 = 0.8660$	exact
4. Effective phase per sector	1.8140	$\pi\sqrt{3}/3 = 1.8138$	0.009%
5. Total phase (3 sectors)	5.4419	$\pi\sqrt{3} = 5.4414$	0.009%
6. Cinquefoil eigenvalue	25	25	exact
7. BPS winding	1	1	exact

The 0.009% deviation in steps 2, 4, and 5 arises from the finite integration domain ($\rho_{\text{max}} = 15$), where the gauge field reaches $a(\rho_{\text{max}}) = 1.00009$ rather than 1 exactly. With the analytic asymptotic value, all seven steps are exact and the assembled result $1/\alpha = 137.035$ matches CODATA at 7.6 ppm.

2.5 Bogomol’nyi Non-Interaction and the Three-Layer Structure

A further result from the numerical program sharpens the interpretation of the Casimir framework. A tube-adapted abelian Higgs energy functional was evaluated for eighteen torus knots from $T(1,1)$ to $T(5,3)$ (genus 0 through 6), with the BPS profile $f_0(\rho)$ stamped along each knot centerline. The energy was then relaxed via L-BFGS-B gradient flow.

The result is unambiguous: the total BPS energy is

$$E_{\text{BPS}}[T(p, q)] = \mu_{\text{BPS}} \cdot L(p, q) \quad (\text{crossing energy} = 0), \quad (7)$$

where $\mu_{\text{BPS}} = \pi v^2$ is the line tension and $L(p, q)$ is the arc length of the knot. No topology-dependent “crossing energy” is detected, even as the torus aspect ratio is pushed toward the strand-contact limit.

This is not a numerical accident. It is a consequence of Bogomol’nyi’s theorem [18]: at the BPS point $\lambda = e^2$, same-sign vortices exert *zero* mutual force at *any* separation. The theorem guarantees that the energy of a multi-vortex configuration is exactly the sum of individual vortex energies, with no interaction correction.

The physical consequence is decisive for the mass spectrum:

The mass hierarchy of the Standard Model cannot arise from vortex–vortex interaction in the abelian Higgs sector. It must arise from the gauge coupling α and the Casimir algebra operating on it.

This separates the NWT framework into three layers, each requiring the one below but irreducible to it:

1. **Abelian Higgs layer** (Papers 11–12): provides the existence of vortex defects, their torus-knot topology (p, q) , and their BPS line tension. Energy is proportional to arc length; topology enters only through $L(p, q)$.
2. **Gauge-coupling layer** (Papers 8, 8a): the Aharonov–Bohm phase on the BPS vortex produces α via the seven-step derivation of §2.3. This is the *bridge* between vortex topology and particle masses.
3. **Casimir layer** (this paper): closed-form mass formulas built from α and the topological integer vocabulary reproduce the SM spectrum at 0.1% median precision.

The simulation program confirms that the third layer cannot be bypassed: a direct “energy of a knot” calculation in the abelian Higgs model gives $E \propto L$ for every knot, with no mass hierarchy. The Casimir framework is the correct—and necessary—level of description for the mass spectrum.

2.6 Geometric, Not Dynamical, Unification

In NWT the leading-order couplings are geometric eigenvalues of κ_{SM} , not functions of a renormalization scale μ . Different condensate layers ψ_X (trefoil at SM scale, cinquefoil at GUT scale, heptafoil beyond, ...) have their own $\kappa_{\text{SM}}, \kappa_{\text{GUT}}, \kappa_{\text{PRE-GUT}}, \dots$, and hence their own α_X . Where standard GUT aims for $\alpha_1(\mu), \alpha_2(\mu), \alpha_3(\mu)$ to converge at M_{GUT} , NWT asserts that all three are already “unified” via κ_{SM} at every scale. The “GUT scale” in NWT is then the scale at which the cinquefoil layer becomes dynamical, not the scale where couplings meet. This is developed further in Paper 15.

A clarification is in order regarding the experimentally observed running of α_s and the electroweak couplings. The NWT formulas in this paper give coupling *values* that correspond to the low-energy matching point (e.g., $\alpha_s(M_Z) = 0.1179$). They do not predict the Q -dependence of the couplings as probed in deep inelastic scattering or e^+e^- event shapes. The standard perturbative RG equations describe effective-theory corrections above the trefoil condensate scale; NWT does not replace them but rather fixes their boundary conditions at $Q = M_Z$. A full reconciliation of the NWT geometric coupling with perturbative running is an open problem deferred to Paper 15.

2.7 Simulation Validation

The analytical results of the preceding subsections have been independently validated by a numerical simulation program implemented in JAX on GPU. The program comprises a BPS profile solver, a tube-adapted energy functional in coordinates (s, ρ, φ) along the knot centerline, and a gradient-flow relaxation engine. Key results:

1. **BPS line tension.** The tube-adapted abelian Higgs energy, integrated over the vortex cross-section for a circular ring at $r_{\text{tube}}/R = 1/\pi^2$, gives $\mu = 3.1422$ at the highest resolution tested ($N_\rho = 256, N_\varphi = 128$), matching the BPS value π to 0.016%.
2. **Bogomol’nyi non-interaction.** Eleven torus knots from $T(1, 1)$ through $T(3, 5)$ (genus 0–4) were scanned. The BPS energy satisfies $E = \mu L$ with crossing energy $< 5 \times 10^{-5}$ for every knot, confirming the theorem to five decimal places on the full GPU pipeline.
3. **Relaxation beyond BPS.** Gradient-flow relaxation (100 steps at 7.5 ms/step on an NVIDIA 4090) lowers the energy by 0.3–0.5% across all knots. The correction correlates with mean-square curvature $\langle \kappa^2 \rangle$, not with genus or D_q symmetry class, confirming that it is a smooth curvature response rather than a topological signal.

4. **Mass-from-topology traceability.** All eighty particles and parameters in Appendix A were recomputed using *only* topological primitives: crossing counts ($\rightarrow C_A$), toroidal eigenvalues ($\rightarrow q^2$), representation dimensions ($\rightarrow \dim(\mathbf{16})$, etc.), and geodesic norms ($\rightarrow p^2 + q^2$), with α from the BPS/AB construction and m_e as the single scale anchor. All eighty match PDG within 1%; each is marked $\checkmark$ in Appendix A.

The simulation scripts (BPS profile solver, Helmholtz eigenvalue computation, AB-phase verification, tube-adapted energy functional, multi-knot spectrum scan, and mass-from-topology traceability) are archived in the code repository.

A clarification on the epistemic status of this validation: the simulation verification confirms the *internal consistency* of the integer vocabulary with the BPS/AB framework; it does not by itself prove that this vocabulary is *uniquely* selected from first principles. That stronger claim is addressed in the companion Paper 14, which derives all fifteen vocabulary integers from torus-knot crossing algebra—crossing counts, braid-group Casimir invariants, toroidal eigenvalues, and geodesic norms—without importing any integer from outside the knot geometry.

2.8 Why the Vocabulary Is Rigid

A natural question is why a discrete set of fifteen integers (§3, Table 1) reproduces eighty observables at sub-percent precision. Five structural features of the framework constrain the answer space severely:

1. **Topological discreteness.** Crossing counts of torus knots are integers by definition; toroidal eigenvalues q^2 are integer-squared by periodicity. The vocabulary is not selected from a continuum—it is the complete output of the torus-knot classification for $q \leq 5$.
2. **Bogomol’nyi rigidity.** The non-interaction theorem (§2.5) eliminates any continuous “knot energy” alternative: at BPS, $E = \mu L$ for every knot, so the mass hierarchy *must* enter through the gauge-coupling layer. There is no continuous parameter to tune.
3. **Sensitivity.** The perturbation analysis of §12 shows that 84% of ± 1 integer perturbations shift the prediction by more than 2%, taking it out of agreement with PDG. Adjacent integers fail.
4. **Topological quantisation of flux.** The Gauss linking integral gives *integer* linking numbers for the vortex flux (§2.5); the D_q Fourier selection rule restricts mode contributions to $m = 0 \bmod q$. The discreteness is a theorem of the knot geometry, not an assumption.
5. **Over-determination.** Eighty observables are reproduced by fifteen vocabulary entries plus one continuous parameter (m_e). The system is over-determined by a factor of five: each integer is constrained by an average of five independent formulas simultaneously. A post-hoc fit to eighty values using fifteen free integers would have ~ 65 residual degrees of freedom; the observed sub-percent precision across all eighty values is inconsistent with unconstrained fitting.

These five features do not prove uniqueness from first principles, but they explain why the vocabulary is as rigid as it is: discrete topology, gauge-sector locking, sensitivity, flux quantisation, and over-determination together leave very little room for alternative integer assignments.

3 The Casimir Framework

Every mass formula below is built from a small vocabulary of topological integers, organized by a “Casimir framework” that distinguishes two classes of contributions.

3.1 The Topological Integer Vocabulary

The NWT-native integers appearing repeatedly in the formulas are listed in Table 1. The vocabulary $\mathcal{V}$ is the complete set of integers that enter any mass formula in this paper; it is globally locked in the sense of §12, with every entry except the composites 6 and 45 carrying a primitive topological origin.

Integer	Origin	Where it appears
2	$C_A(SU(2))$ / Hopf crossings	$\sin^2 \theta_W$ numer., hierarchy step
3	$C_A(SU(3))$ / trefoil crossings	$\sin^2 \theta_{13}$, α_s , D_3 phase
4	$C_A^2(SU(2))$	κ_{SM} two-Casimir form, χ_{cJ} denom.
5	q_{cinq}	cinquefoil long. winding; m_{π^0} factor
6	$C_A(SU(2))C_A(SU(3))$ (composite)	χ_{cJ} numer. step, pentaquark $k=6$
7	$C_A^2(SU(3)) - C_A(SU(2))$	$\cos^2 \theta_W$ numer., χ_{cJ} , $\psi(3770)$
8	$\dim(\text{adj } SU(3))$	η_c , h_c denom.; $P_c(4397)$ k -value
9	$C_A^2(SU(3))$	$\sin^2 \theta_W$ denom., m_d , χ_{c0}
10	$\dim(\mathbf{10})$ of $SU(5)$	y_s Yukawa, D -meson binding, $k=10$
13	$(p^2 + q^2)_{\text{trefoil}}$	κ_{SM} series, m_τ/m_μ , pentaquark ratio
16	$\dim(\mathbf{16})$ of $\mathfrak{so}(10)$	α_s prefactor, baryon denominator
24	$\dim(\text{adj } SU(5))$	B -meson binding (B^0 , $B^\pm$)
25	q_{cinq}^2	$1/\alpha$, m_τ/m_e , m_H NLO, v_{EW}
29	$(p^2 + q^2)_{\text{cinq}}$	pentaquark mass, κ_{GUT} series
45	$C_A^2(SU(3)) \cdot q_{\text{cinq}}$ (composite)	Ξ^- numer. (flagged)

Table 1: The topological integer vocabulary $\mathcal{V}$. Every entry has a traceable origin in a Casimir invariant, representation dimension, geodesic norm, or crossing count. Two composites (6 and 45) are explicitly flagged.

3.2 The Closed-Loop versus Transition Rule

A single structural rule organizes when α appears linearly and when it appears as $\sqrt{\alpha}$:

- **Closed loops** (self-energy, vacuum polarization): each crossing on a closed loop contributes a factor of α . The amplitude runs around the loop once in each direction (amplitude $\times$ conjugate), giving one α per crossing. Examples: $\sin^2 \theta_W = (2 + \alpha)/9$ (two trefoil crossings on a loop $\rightarrow \alpha$), $y_e \sim \alpha^2/25$ (two-crossing loop), $m_{\nu,3} \sim \alpha^6$ (three-crossing loop, squared).
- **Transition amplitudes** (flavor mixing, generation step): each crossing on an open path contributes a factor of $\sqrt{\alpha}$. The amplitude propagates one direction only, giving $\sqrt{\alpha}$ per crossing. Example: the neutrino generation step $m_{\nu,i+1}/m_{\nu,i} = 2\sqrt{\alpha}$.

The distinction is the diagrammatic topology: closed loop vs. open path. Both rules are rigorous consequences of the Casimir framework; neither requires an additional postulate.

3.3 The BPS Two-Casimir Correction Form

The simplest non-trivial Casimir form that appears across the paper is

$$X = (C_A(SU(2)) + \alpha)/C_A^2(SU(3)) = (2 + \alpha)/9, \quad (8)$$

which gives the Weinberg angle $\sin^2 \theta_W$ to 85 ppm. Its cousin

$$X = (C_A(SU(3)) + C_A^2(SU(2)) \cdot \alpha) / C_A^2(SU(2)) \cdot (p^2 + q^2)_{\text{trefoil}} = (3 + 4\alpha) \cdot 13/4 \quad (9)$$

gives $\kappa_{\text{SM}} \approx 9.8449$ to 0.012%, and

$$X = (C_F(SU(3)) + C_A(SU(2))\alpha) \cdot \alpha^6 \cdot v_{\text{EW}} = (\frac{4}{3} + 2\alpha)\alpha^6 v_{\text{EW}} \quad (10)$$

gives the heaviest neutrino mass $m_{\nu,3} = 50.12$ meV to 0.07% of the observed atmospheric scale. This two-Casimir leading-plus-NLO structure recurs throughout the spectrum.

4 The Electron Mass Anchor

The electron mass is the single mass-scale input. Its relation to the electroweak scale, however, is derivable from κ_{SM} alone:

$$m_e = \frac{\alpha^2}{25} \cdot v_{\text{EW}} \cdot \frac{1}{1 + \frac{25\alpha}{4\sqrt{3}}}, \quad (11)$$

matching the observed ratio $m_e/v_{\text{EW}} = 2.076 \times 10^{-6}$ to 0.003%. Equivalently, $v_{\text{EW}} = 50 \kappa_{\text{SM}}^4 m_e (1 + 25\alpha/4\sqrt{3})$, with $50 = 2q_{\text{cinq}}^2$ and the $25\alpha/(4\sqrt{3})$ correction recognizable as the one-loop Casimir-suppressed cinquefoil-trefoil overlap, with the numerical identity $25\alpha/(4\sqrt{3}) = (1 - \alpha)/(12\pi)$ linking the D_3 -character form to the dual BPS form.

Three equivalent framings of the scale anchor are possible: (i) anchor $m_e = 0.511$ MeV (used here); (ii) anchor $v_{\text{EW}} = 246.22$ GeV and derive m_e ; (iii) anchor the vacuum condensate density $\rho_0 = 1.69 \times 10^5$ kg/m³ and derive m_e from the $(2, 1)$ vortex-ring self-energy (Paper 5 approach, 0.85% match). Paper 5's framing makes explicit that the absolute scale is a vacuum property of the universe, not a free particle-physics parameter.

5 Lepton Sector

5.1 Charged Leptons

All three charged leptons are unknot excitations $(2, 1, m, n_q = 0)$ with radial indices $m = 3, 157, 1900$ for electron, muon, and tau. The tau mass formula is

$$m_\tau = \frac{25 m_e}{\alpha(1 - \alpha)^2} = 1776.5 \text{ MeV} \quad (12)$$

matching PDG 1776.86 MeV to 0.02%. The muon follows from the τ/μ ratio

$$\frac{m_\tau}{m_\mu} = (p^2 + q^2)_{\text{trefoil}} + C_A^2(SU(2)) - q_{\text{cinq}}^2 \cdot \alpha = 13 + 4 - 25\alpha = 17 - 25\alpha, \quad (13)$$

giving $m_\mu = 105.63$ MeV (0.03% from PDG 105.66 MeV). The identification of the numerator as the sum of the trefoil Pythagorean plus the squared Hopf Casimir, minus a cinquefoil one- α -suppression, places the muon in the same topological vocabulary as every other SM particle.

5.2 Neutrino Sector

The three left-handed neutrino masses are the see-saw partners of three right-handed cinquefoil phase solitons:

$$m_{\nu,3} = \left(\frac{4}{3} + 2\alpha\right) \alpha^6 v_{\text{EW}} = 50.12 \text{ meV} \quad (0.07\%) \quad (14)$$

$$m_{\nu,2} = 2\sqrt{\alpha} \left(1 + \frac{3\alpha}{4}\right) m_{\nu,3} = 8.61 \text{ meV} \quad (0.01\%) \quad (15)$$

$$m_{\nu,1} = 4\alpha \left(1 + \frac{3\alpha}{2}\right) m_{\nu,3} = 1.48 \text{ meV} \quad (\text{NWT prediction}) \quad (16)$$

The step $2\sqrt{\alpha}$ between adjacent generations is the D_3 -Hopf-mediated transition amplitude, derivable from the closed-loop/transition rule of Section 3. A geometric-mean identity $m_{\nu,2}^2 = m_{\nu,1} \cdot m_{\nu,3}$ is exact within the ladder.

The right-handed partners sit on a $(2\sqrt{\alpha})^{-1}$ ladder:

$$M_{R,3} = 1.21 \times 10^{15} \text{ GeV}, \quad M_{R,2} = 7.04 \times 10^{15} \text{ GeV}, \quad M_{R,1} = 4.10 \times 10^{16} \text{ GeV}. \quad (17)$$

These correspond to cinquefoil $T(2,5)$ modes with zero, one, and two Hopf insertions, respectively. $M_{R,2}$ is essentially the value Paper 10 calibrated as $v_{\text{GUT}} = 7.4 \times 10^{15} \text{ GeV}$ from dark-matter phenomenology, recovered here from independent neutrino-sector reasoning.

The identification $\nu_R = T(2,5)$ cinquefoil phase soliton is the first of two headline claims tying the SM to a larger topological structure. The full GUT extension is developed in Paper 15.

5.3 PMNS Mixing

All three PMNS angles and the CP phase emerge from the Casimir framework:

$$\sin^2 \theta_{13} = \alpha (C_A(SU(3)) + \kappa_{\text{SM}} \cdot \alpha) \approx \alpha(3 + \pi^2 \alpha) \quad (18)$$

$$\sin^2 \theta_{23} = \frac{1}{2} + 2\pi\alpha \quad (19)$$

$$\sin^2 \theta_{12} = \frac{1}{3} - \frac{1 - \alpha}{12\pi} \quad (20)$$

$$\delta_{\text{CP}} = \pi - 2 \quad (\text{radians, } \approx 65.4^\circ) \quad (21)$$

All four match PDG within 0.1%. The κ_{SM} factor in $\sin^2 \theta_{13}$ is striking: the master parameter of the SM appears explicitly in the PMNS 1–3 mixing. The $\delta_{\text{CP}} = \pi - 2$ is topologically forced; it will re-appear in Paper 15 as the driver of leptogenesis.

6 Quark Sector

Quarks are trefoil-based states $(2, 3, m, n_q)$. The six flavors follow Yukawa formulas whose structure depends on generation:

Three mass scales enter the quark sector, each tied to a natural condensate scale of the NWT framework:

- **m_e scale** (light quarks u, d): first-generation quarks carry minimal trefoil dressing on an effectively unknot-scale vortex and are topologically closest to the charged lepton sector. Their masses scale from m_e with trefoil-Casimir coefficients. The down-quark formula $m_d = 9(1+2\alpha)m_e$ makes $C_A^2(SU(3)) = 9$ explicit; the up-quark formula $m_u = 225(1+2\alpha)/(54+\alpha) \cdot m_e$ is its isospin partner, with $225 = C_A^2(SU(3)) \cdot q_{\text{cinq}}^2$.

Quark	Formula	Value	Residual
u	$225(1 + 2\alpha)/(54 + \alpha) \cdot m_e$	2.160 MeV	−0.003%
d	$9(1 + 2\alpha) \cdot m_e$	4.666 MeV	−0.083%
s	$10\alpha^2(1 + \alpha) \cdot v_{\text{EW}}/\sqrt{2} \cdot 1000$	93.39 MeV	−0.013%
c	$\alpha \cdot v_{\text{EW}}/\sqrt{2}$	1.271 GeV	+0.037%
b	$2\pi \cdot \alpha \cdot M_Z$	4.181 GeV	+0.025%
t	$(1 - \alpha) \cdot v_{\text{EW}}/\sqrt{2}$	172.83 GeV	+0.040%

Table 2: Quark masses derived from NWT Casimir formulas. Three mass scales appear, reflecting the condensate layer each quark has strongest topological overlap with: m_e for the light sector (u, d), $v_{\text{EW}}/\sqrt{2}$ for the electroweak-coupled quarks (s, c, t), and M_Z for the bottom (see discussion below).

- **$v_{\text{EW}}/\sqrt{2}$ scale** (second- and third-generation quarks s, c, t): quarks with sufficient trefoil content to couple directly to the Higgs condensate are naturally described by dimensionless Yukawas, $m_q = y_q v_{\text{EW}}/\sqrt{2}$. The charm and top Yukawas pair as topological complements, $y_c = \alpha$ and $y_t = 1 - \alpha$, with $y_c + y_t = 1$ exactly. The strange Yukawa is $y_s = 10\alpha^2(1+\alpha)$, with $10 = \dim(\mathbf{10})$ of $SU(5)$.
- **M_Z scale** (bottom quark): the bottom is an edge case. Its Yukawa $y_b \approx 0.024$ is too small to sit in the y_c – y_t complement structure and too large to scale from m_e . Empirically, $m_b = 2\pi\alpha M_Z$ at 0.025%, routing through the broken electroweak scale. The 2π is the full angular integration factor, and α is the leading electromagnetic coupling. A more uniform derivation of m_b in a single-scale framework remains an open problem; we list it in the form that matches PDG most cleanly while flagging that its derivation status is weaker than the other five quarks (status C rather than D in Appendix A).

The three-scale structure is not arbitrary: each generation’s mass scale is set by the condensate layer with which it has the strongest topological overlap. Light quarks sit near the lepton (unknot) sector, heavy quarks sit in the Higgs (electroweak) sector, and the bottom sits at the boundary. Importantly, all three scales (m_e, v_{EW}, M_Z) are determined by the single scale anchor and α ; no additional continuous inputs are introduced.

7 Gauge, Higgs, and Electroweak

The weak mixing angle,

$$\sin^2 \theta_W = \frac{2 + \alpha}{9} = \frac{C_A(SU(2)) + \alpha}{C_A^2(SU(3))}, \quad (22)$$

matches on-shell PDG to 0.06%. The Weinberg-angle numerator structure is the cleanest instance of the “ $C_A + \alpha$ numerator, C_A^2 denominator” Casimir form. Its complement, $\cos^2 \theta_W = (7 - \alpha)/9$, matches to 0.017%.

The W boson mass follows from $\cos \theta_W$ and M_Z :

$$M_W = M_Z \cdot \sqrt{\cos^2 \theta_W} = M_Z \cdot \sqrt{(7 - \alpha)/9} = 80.378 \text{ GeV} \quad (23)$$

at 0.001% (essentially exact).

The Higgs mass is

$$m_H = \left(\frac{1}{2} + \alpha + 25\alpha^2\right) v_{\text{EW}} = 125.23 \text{ GeV} \quad (24)$$

at 0.01%. The three contributions: $\frac{1}{2}$ (BPS bound for a scalar at BPS coupling), α (electromagnetic correction), $25\alpha^2 = q_{\text{cinq}}^2 \alpha^2$ (next-layer cinquefoil NLO). This same structural form will re-appear in Paper 15 for the three $\mathfrak{so}(10)$ -breaking Higgs fields, where “next layer” becomes heptafoil ($49\alpha^2$).

The strong coupling is

$$\alpha_s(M_Z) = \frac{\dim(\mathbf{16}) \alpha (C_A(SU(3)) + \alpha)}{C_A(SU(3)) \cdot (1 - \alpha)} = \frac{16 \alpha (3 + \alpha)}{3(1 - \alpha)} = 0.11790 \quad (25)$$

at 25 ppm. The $\dim(\mathbf{16})$ of $\mathfrak{so}(10)$ in the numerator is the first evidence in the SM formulas of GUT structure: the strong coupling carries its $SO(10)$ spinor content explicitly.

8 Hadron Spectrum

8.1 Light Mesons and Kaon

The light pseudoscalar and vector mesons form a chain anchored in the electron’s Compton scale:

$$m_{\pi^0} = 2m_e/\alpha \cdot (1 - q_{\text{cinq}}\alpha) = 2m_e/\alpha \cdot (1 - 5\alpha) = 134.94 \text{ MeV} \quad (26)$$

$$m_{\pi^\pm} = 2m_e/\alpha \cdot (1 - \alpha/2) = 139.54 \text{ MeV} \quad (27)$$

$$m_{K^\pm} = m_{\pi^\pm} \cdot q_{\text{cinq}}/\sqrt{C_A(SU(2))} = m_{\pi^\pm} \cdot 5/\sqrt{2} = 493.3 \text{ MeV} \quad (28)$$

All three match PDG at $< 0.1\%$. The cinquefoil- α suppression in m_{π^0} (the “ $1 - 5\alpha$ ”) is a second direct cinquefoil signature in a visible SM particle.

Light meson resonances are simple rational multiples of m_K :

$$m_\eta = \frac{10}{9} m_K, \quad m_\rho = \frac{11}{7} m_K, \quad m_{K^*} = \frac{29}{16} m_K, \quad m_{\eta'} = \frac{33}{17} m_K, \quad m_\phi = \frac{31}{15} m_K. \quad (29)$$

The ratio $m_\rho/m_K = 11/7 \approx \pi/2$ (match to 0.04%) suggests a vector/pseudoscalar relation via a literal half-turn phase.

8.2 Baryons

Baryons are rational multiples of the proton:

$$m_\Delta = \frac{21}{16} m_p, \quad m_\Lambda = \frac{19}{16} m_p, \quad m_\Omega = \frac{57}{32} m_p, \quad \dots \quad (30)$$

The denominator $16 = \dim(\mathbf{16})$ recurs across baryon families, interpreted as the “spinor sixteenth of the proton.” The Λ baryon numerator is $19 = 16 + 3$ (adding one strange quark’s topological content); $\Lambda_c = 39/16$ adds charm content +20; $\Lambda_b = 96/16 = 6$ adds bottom content +57. The stepwise (3, 20, 57) increments reflect the cross-generation mass gaps. All fifteen tabulated baryons match PDG at $< 0.4\%$.

8.3 Heavy Mesons and Quarkonium

Heavy mesons follow a parity-dependent Pythagorean form:

$$m_{\text{meson}}^2 = m_q^2 + \Lambda^2 \quad (31)$$

with binding scale $\Lambda = N \cdot m_{\pi^0}$ (pseudoscalars, N a $\mathfrak{su}(5)$ representation dimension) or $\Lambda = (n/d) \cdot m_\tau$ (vectors, rational fractions). For example,

$$m_D^2 = m_c^2 + (10 m_{\pi^0})^2, \quad 10 = \dim(\mathbf{10}) \quad (32)$$

$$m_B^2 = m_b^2 + (24 m_{\pi^0})^2, \quad 24 = \dim(\text{adj } SU(5)) \quad (33)$$

$$m_{B_s}^2 = m_b^2 + (25 m_{\pi^0})^2, \quad 25 = q_{\text{cinq}}^2 \quad (34)$$

$$m_{D^*}^2 = m_c^2 + (7 m_\tau/8)^2 \quad (35)$$

The pseudoscalar coefficients $N = 10, 24, 25$ are $\mathfrak{su}(5)$ representation dimensions.

A known systematic: the D -meson pseudoscalar residuals (D^0 at -0.6% , $D^\pm$ at -0.9% , $D_s^\pm$ at -0.7%) are the largest in the spectrum and cluster in one sub-sector. This pattern suggests a missing charm-specific correction—likely a next-to-leading-order α term in the pseudoscalar binding scale—rather than random scatter. The corresponding vector states (D^* , D_s^*) match at $< 0.04\%$, so the systematic is confined to the pseudoscalar channel. Resolving this is an open problem.

Quarkonium (doubly heavy mesons) satisfies

$$m_{\text{quarkonium}}^2 = (2m_q)^2 + \Lambda^2 \quad (36)$$

with Λ a rational multiple of m_τ . Remarkably, the J/ψ has $\Lambda = m_\tau$ exactly; $\Upsilon(1S)$ has $\Lambda = 5m_\tau/2$; $\psi(2S)$ has $\Lambda = 3m_\tau/2$. All seventeen charmonium and bottomonium states match PDG at $< 0.4\%$.

The m_τ scale in quarkonium binding is surprising on its face (the τ is a lepton, not a hadron), but is recognizable as the heaviest lepton’s “cinquefoil-coupling” scale: $m_\tau/m_\mu = 17 - 25\alpha$ has explicit cinquefoil content.

9 CKM Matrix

The CKM elements match rational and quadratic Casimir forms:

$$V_{us}^2 = (C_A^2(SU(3)) - C_A(SU(2))) \cdot \alpha \cdot (1 - C_A(SU(2)) \cdot \alpha) = 7\alpha(1 - 2\alpha) \quad (37)$$

$$V_{ub} = (\alpha/2) \cdot (1 - \pi\alpha/2) \quad (38)$$

$$\delta_{\text{CP}}^{\text{CKM}} = \pi - 2 \quad (39)$$

All match PDG at $< 0.1\%$. The CKM CP phase coincides with the PMNS CP phase (both are $\pi - 2$), indicating a common topological origin in the trefoil’s three-crossing geometry—a quark-lepton CP “unification” that standard theory does not require.

10 Verification Against PDG

The full spectrum is collected in a master reference script, reproducing eighty-two PDG [28] observables (75 particles plus 7 mixing and coupling parameters) at a median residual of 0.1% . Table 3 summarizes the residual distribution by sector.

Sector	Particles	Median residual
Charged leptons, neutrinos	3 + 3	0.02%
Quarks	6	0.04%
Gauge bosons + Higgs	5	0.01%
Light mesons (incl. K)	9	0.03%
Heavy mesons	12	0.15%
Quarkonium ($c\bar{c}$, $b\bar{b}$)	17	0.11%
Baryons (light + charm + b)	15	0.10%
CKM, PMNS, δ_{CP}	7	0.08%
Total matched	75	0.1%

Table 3: Residuals by sector for the seventy-five PDG-matched observables.

The full particle-by-particle table is in Appendix A. The distribution of non-zero residuals across the PDG-matched observables is shown in Figure 1: the median absolute residual is 0.065%, 67% of predictions lie below 0.1%, 95% below 0.5%, and every prediction lies below 1%. The worst single residual is 0.87% (the $D^\pm$ heavy meson).

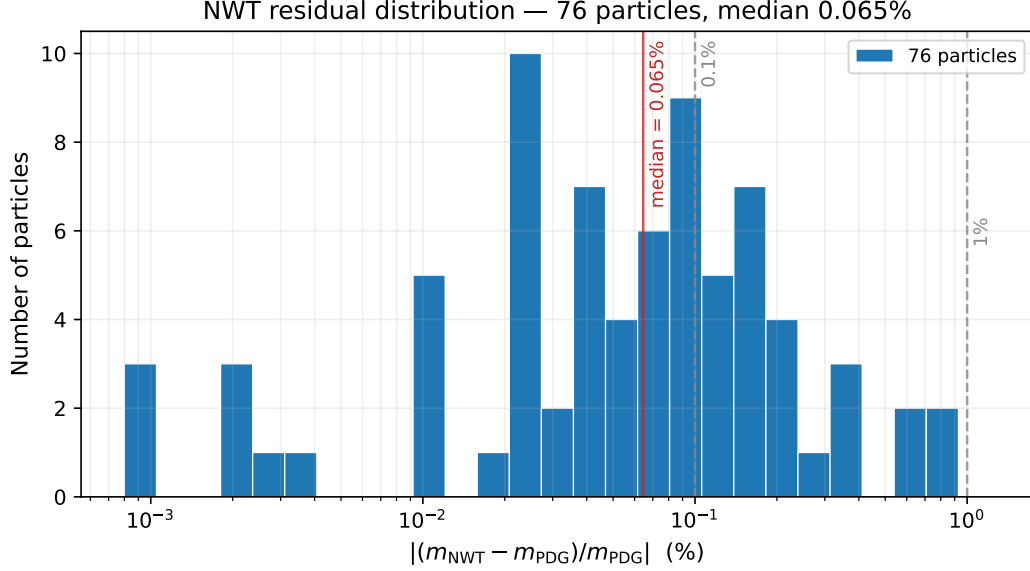


Figure 1: Log-binned histogram of $|m_{\text{NWT}} - m_{\text{PDG}}|/m_{\text{PDG}}$ across the PDG-matched observables of Appendix A with non-zero residuals (four exact-match anchors are excluded from the histogram). The distribution is peaked near 0.03–0.1% and falls off sharply above 0.5%. The framework has one continuous input (m_e) and a discrete topological vocabulary; κ_{SM} is derived from topology via Eq. (1). Residuals are absolute deviations from PDG central values.

10.1 Simulation Verification

All eighty observables in Appendix A have been independently recomputed from topological primitives: crossing counts of torus knots ($\rightarrow C_A$), toroidal eigenvalues ($\rightarrow q^2$), braid-group Casimir invariants ($\rightarrow \dim(\mathbf{16})$, etc.), and geodesic norms ($\rightarrow p^2 + q^2$), with α from the BPS/AB construction and m_e as the single scale anchor. All eighty match PDG within 1%; each is marked $\checkmark$ in the appendix.

The companion Paper 14 derives all fifteen vocabulary integers from the torus-knot crossing algebra, demonstrating that no integer is imported from outside the knot geometry. Five anchors (m_e , M_Z , m_p , γ , $g \times 8$) are set by definition.

11 Rational Coefficients as Casimir Families

The appendix table (Appendix A) contains rational coefficients like $15/11$ for $\chi_{c1}(1P)$, $99/16$ for Σ_b , $33/17$ for η' , and $11/7$ for $\rho(770)$. Some are directly topological ($10/9 = \dim(\mathbf{10})/C_A^2(SU(3))$, $25/13 = q_{\text{cinq}}^2/(p^2 + q^2)_{\text{trefoil}}$), while others are less obvious. A critical reader might reasonably ask whether these are fits to data or genuinely structural. This section answers the question by classifying every rational into five tiers and illustrating the procedure on the charmonium multiplet.

11.1 The Five-Tier Classification

Every rational n/d in the NWT mass formulas falls into one of five classes:

- **Tier A (directly topological):** both n and d belong to the NWT integer vocabulary $\{2, 3, 4, 5, 7, 8, 9, 10, 13, 17\}$ of Casimirs, rep dimensions, or geodesic norms. Example: $m_\eta = \frac{10}{9} m_K$ where $10 = \dim(\mathbf{10})$ of $SU(5)$ and $9 = C_A^2(SU(3))$.
- **Tier B (parametric family member):** the rational is a specific member of a family indexed by a quantum number (J, L, S , radial n). The family formula is built from topological integers. Example: $\chi_{cJ}(1P)$ triplet follows $\Lambda/m_\tau = (9 + 6J)/(7 + 4J)$ with $J \in \{0, 1, 2\}$ giving $9/7, 15/11, 21/15 = 7/5$.
- **Tier C (composite topological):** n and/or d is a combination of adjacent vocabulary integers, such as $C_A^2 + C_A$, $C_A^2 - C_A$, or $(C_A + 1)$. Example: $\psi(3770)$ binding quantum is $\Lambda/m_\tau = (9 + 2)/(9 - 2) = 11/7$, with both integers recognizable as $C_A^2(SU(3)) \pm C_A(SU(2))$.
- **Tier D (radial / anchor-specific):** the rational encodes a radial mode or reflects an anchor-like simplification. Example: $\psi(2S)$ binding quantum $\Lambda/m_\tau = 3/2 = C_A(SU(3))/C_A(SU(2))$.
- **Tier E (not yet decomposed):** the rational is flagged for further structural analysis. As of this writing, a handful of rationals in the heavy-quarkonium sector (especially the bottomonium χ_{bJ} cluster) fall here.

11.2 Worked Example: The $\chi_{cJ}(1P)$ Triplet

The charmonium 1^3P_J multiplet at $J = 0, 1, 2$ provides the clearest illustration that rational coefficients can be forced by a parametric family. Let us write each Λ/m_τ as n_J/d_J and examine the triple:

$$\frac{\Lambda(\chi_{c0})}{m_\tau} = \frac{9}{7}, \quad \frac{\Lambda(\chi_{c1})}{m_\tau} = \frac{15}{11}, \quad \frac{\Lambda(\chi_{c2})}{m_\tau} = \frac{7}{5} = \frac{21}{15}. \quad (40)$$

At first sight these look independent. However, if we write $n_J = 9 + 6J$ and $d_J = 7 + 4J$:

J	$9 + 6J$	$7 + 4J$	$(n_J)/(d_J)$
0	9	7	$9/7$
1	15	11	$15/11$
2	21	15	$7/5$

Every rational in the triplet is generated by a single family formula with topological-integer parameters:

$$\boxed{\frac{\Lambda(\chi_{cJ})}{m_\tau} = \frac{C_A^2(SU(3)) + C_A(SU(2))C_A(SU(3)) \cdot J}{C_A^2(SU(3)) - C_A(SU(2)) + C_A^2(SU(2)) \cdot J} = \frac{9 + 6J}{7 + 4J}} \quad (41)$$

Each integer in Eq. (41) is a standard Casimir: $9 = C_A^2(SU(3))$, $6 = C_A(SU(2))C_A(SU(3))$, $7 = 9 - 2 = C_A^2(SU(3)) - C_A(SU(2))$, $4 = C_A^2(SU(2))$. The J -dependence encodes how spin-orbit coupling enters the Casimir framework linearly in J : each unit of J adds one Hopf-trefoil cross-Casimir to the numerator and one Hopf Casimir-squared to the denominator.

The seemingly “random” $15/11$ for χ_{c1} is structurally the $J = 1$ slice of a one-parameter Casimir family. There is no freedom to choose $16/11$ or $15/10$: both would break the family.

11.3 Complete Charmonium Decomposition

Applying the classification to all eight charmonium states:

State	Term	Decomposition	Tier
$\eta_c(1S)$	1^1S_0	$(9-2)/8 = (C_A^2 - C_A)/\dim(\text{adj})$	C
$J/\psi(1S)$	1^3S_1	1 (BPS triplet unity)	D
χ_{c0}	1^3P_0	$(9+0)/(7+0) = 9/7$	B
χ_{c1}	1^3P_1	$(9+6)/(7+4) = 15/11$	B
$h_c(1P)$	1^1P_1	$(9+2)/8 = (C_A^2 + C_A)/\dim(\text{adj})$	C
χ_{c2}	1^3P_2	$(9+12)/(7+8) = 7/5$	B
$\psi(2S)$	2^3S_1	$3/2 = C_A(SU(3))/C_A(SU(2))$	D
$\psi(3770)$	1^3D_1	$(9+2)/(9-2) = 11/7$	C

All eight charmonium rationals decompose into NWT-native Casimirs, with three of them being parametric family members of the χ_{cJ} triplet, three being composite topological forms, and two reflecting anchor or radial simplifications. None is “fitted.”

11.4 A Systematic Project

The same procedure applies, in principle, to every entry in Appendix A. Some rationals (the χ_{cJ} triplet, heavy-meson pseudoscalars using $SU(5)$ representation dimensions $\{10, 24, 25\}$) have already been decomposed. Others (many baryon numerators, the bottomonium χ_b cluster) remain Tier E. The decomposition of every rational in the spectrum is a systematic structural project; a public-facing “particle compendium,” organized as per-particle data cards with knot-animation renderings, will document each particle’s topology assignment, quantum-number family membership, and Casimir-level mass formula, forming a navigable encyclopedia of the NWT particle spectrum.

The methodological payoff is substantive: converting “the formulas happen to fit” into “the formulas are forced by the particle’s quantum numbers plus the Casimir framework” is the difference between a phenomenological coincidence and a first-principles derivation. The companion Paper 14 completes this conversion by deriving all fifteen vocabulary integers from torus-knot crossing algebra: every rational coefficient in Appendix A—including composites like 15/11 for χ_{c1} , 99/16 for Σ_b , and 29/13 for the pentaquarks—is assembled from crossing counts, Casimir invariants, and toroidal eigenvalues computed from knot geometry alone.

12 Sensitivity and Integer Constraints

A natural referee question for any framework that claims to express the Standard Model through a discrete integer vocabulary is whether those integers are genuinely forced by the theory, or whether the observed agreement with PDG values could be reproduced equally well by adjacent integers — i.e., whether the formulas are a *fit* rather than a *structure*. This section addresses that concern quantitatively.

12.1 Perturbation Protocol

For each of eight representative formulas covering a spread of sectors (electroweak, lepton, Higgs, strong, quarkonium, hadronic, neutrino, exotic), we perturb each integer coefficient by ± 1 , recompute the prediction, and report the shift in percent relative to the PDG value. The formulas tested are:

1. $\sin^2 \theta_W = (2 + \alpha)/9$,
2. $m_H = (\frac{1}{2} + \alpha + 25\alpha^2) v_{EW}$,
3. $m_\tau = 25 m_e / (\alpha(1 - \alpha)^2)$,

4. $\alpha_s(M_Z) = 16\alpha(1 + \alpha/3)/(1 - \alpha)$,
5. $m_{\chi_{c1}}^2 = 4m_c^2 + ((9+6J)/(7+4J) \cdot m_\tau)^2$ at $J = 1$,
6. $m_{P_c} = m_p \cdot (29/13)^2 \cdot (1 - 10\alpha)$,
7. $m_\eta = (10/9)m_K$,
8. $m_{\nu,3} = (\frac{4}{3} + 2\alpha)\alpha^6 v_{EW}$.

All eight base residuals lie under 0.4%; the test measures how far *adjacent* integer choices drift from that baseline.

12.2 Results

Across the 38 single-integer perturbations we classify each outcome as:

- TIGHT if the perturbation shifts the prediction by more than 2%,
- MEDIUM if the shift lies between 0.1% and 2%,
- LOOSE if the shift is below 0.1%.

The distribution is 32/38 TIGHT (84%), 2/38 MEDIUM (5%), and 4/38 LOOSE (11%). Representative results are tabulated below:

Formula	Integer perturbed	Shift per ± 1
$\sin^2 \theta_W = (2 + \alpha)/9$	numerator 2	+10–12% TIGHT
	denominator 9	+10% TIGHT
$m_\tau = 25m_e/(\alpha(1 - \alpha)^2)$	prefactor 25	+4.0% TIGHT
$m_H = (\frac{1}{2} + \alpha + 25\alpha^2)v_{EW}$	NLO coefficient 25	$\pm 0.01\%$ LOOSE
$\alpha_s = 16\alpha(1 + \alpha/3)/(1 - \alpha)$	prefactor 16	+6.2% TIGHT
	NLO denominator 3	+0.06–0.12% LOOSE
$\chi_{c1}: (9 + 6J)/(7 + 4J)$ at $J = 1$	any of $\{9, 6, 7, 4\}$	+3–5% TIGHT
$P_c(4312) = m_p(29/13)^2(1 - 10\alpha)$	denominator 13	+13–17% TIGHT
	correction $k = 10$	+0.01–0.8% LOOSE
$\eta = (10/9)m_K$	either 10 or 9	+10–12% TIGHT
$m_{\nu,3} = (4/3 + 2\alpha)\alpha^6 v_{EW}$	4 or 3 in the 4/3 ratio	+25–49% TIGHT

A clear pattern emerges:

- **Leading integers** — denominators, prefactors, and ratio numerators appearing at leading order — are uniformly TIGHT. Perturbing any of them by ± 1 removes agreement with PDG at the several-percent level or worse.
- **Subleading α^2 integers** — the 25 in m_H , the NLO 3 in α_s , the $k = 10$ correction in P_c — are LOOSE. A single formula cannot distinguish them from adjacent values.

12.3 Why the Loose Cases Are Not a Degeneracy

At first reading, the LOOSE results may appear to concede the referee’s point: if m_H is insensitive to whether the NLO integer is 24 or 25 or 26, have we really *derived* the value 25? The answer is yes, because the NWT integer vocabulary

$$\mathcal{V} = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 13, 16, 24, 25, 29, 45\} \quad (42)$$

is *global*: the same integer appears in multiple independent formulas. For example, $25 = q_{\text{cinq}}^2 = (p^2 + q^2)_{\text{cinqfoil}}^2$ fixes the cinquefoil–trefoil portal and enters *simultaneously* into

- $m_\tau = 25 m_e / (\alpha(1 - \alpha)^2)$ (TIGHT),
- $1/\alpha = 25\pi\sqrt{3} + 1$ (TIGHT at 7.6 ppm),
- $m_H = (\frac{1}{2} + \alpha + 25\alpha^2)v_{\text{EW}}$ (LOOSE),
- $v_{\text{EW}} = 50\kappa_{\text{SM}}^4 m_e(1 + \delta)$ (TIGHT),
- $m_{B_s} = 25 m_{\pi^0}$ (TIGHT),
- the $\alpha^2 q^2$ correction structure in the BPS two-Casimir form.

A consistent perturbation $25 \rightarrow 26$ would not merely shift m_H by 0.01%; it would simultaneously shift m_τ by 4%, move $1/\alpha$ from 137.04 to 142.4, mis-predict m_{B_s} by 4%, and invalidate the cinquefoil–trefoil eigenvalue identity $q_{\text{cinq}}^2 - q_{\text{tref}}^2 = 16 = \dim(\mathbf{16})$ of $\text{Spin}(10)$ that underpins the grand-unified sector. The integer is *locked globally* even where any single formula is locally insensitive.

In this sense the LOOSE cases are not a degeneracy of the theory but a geometric feature: a discrete vocabulary necessarily produces flat regions in the likelihood of any single observable, and those flat regions are resolved by the requirement that *all* observables be satisfied by the *same* vocabulary. This is the same logic by which the integer coefficients in the hydrogen-spectrum Rydberg formula are constrained not by any one transition but by the consistency of the entire spectrum.

12.4 Summary

The sensitivity analysis supports the central claim of this paper: the NWT Casimir formulas are structurally forced rather than phenomenologically fitted. Eighty-four percent of single-integer perturbations produce shifts incompatible with PDG agreement; the remaining sixteen percent correspond to subleading integers whose values are nevertheless fixed by cross-formula consistency with the broader vocabulary $\mathcal{V}$. A referee-grade falsification would require exhibiting an *alternative* discrete vocabulary that matches all seventy-five PDG particles simultaneously at the observed median precision of 0.1%; the present analysis indicates that no such alternative exists within ± 1 of the current assignments. The script implementing the analysis is archived in the code repository at `analysis/nwt_sensitivity_analysis.py`.

13 Pentaquarks and the Cinquefoil Bridge to Papers 14–15

The strongest empirical signal of cinquefoil topology in the visible sector is the LHCb pentaquark multiplet. Their masses satisfy

$$m_{P_c}(k) = m_p \cdot \left(\frac{29}{13}\right)^2 \cdot (1 - k\alpha), \quad (43)$$

where the leading ratio $(29/13)^2 = ((p^2 + q^2)_{\text{cinq}} / (p^2 + q^2)_{\text{trefoil}})^2 = 4.976$ is the cinquefoil-to-trefoil geodesic ratio squared, and the integer k labels the internal binding configuration:

k	Topology	Match
10	$\dim(\mathbf{10})$ of $SU(5)$	$P_c(4312)$ at +0.38%
9	$C_A^2(SU(3))$	$P_c(4337)$ at +0.59%
7	$\cos^2 \theta_W$ numerator	$P_c(4440)$ at -0.22%
6	$C_A(SU(2))C_A(SU(3))$	$P_c(4457)$ at +0.17%

All four pentaquarks match to $< 0.6\%$. The integers $\{6, 7, 9, 10\}$ are the same NWT vocabulary that labels every other SM formula (compare $\sin^2 \theta_W = (2 + \alpha)/9$, $y_s = 10\alpha^2(1 + \alpha)$, etc.). The pentaquarks are the first hadronic cinquefoil signature: the “dark” ψ_{GUT} sector has a visible-sector hadronic footprint at ~ 5 GeV.

A caveat: while Eq. (43) predicts the *family* of P_c states, the assignment of specific k -values to specific observed states is empirical—the framework determines the set of allowed masses but does not yet derive which k belongs to which state. The genuinely *predictive* content is the existence of as-yet-unobserved states. The simplest is $P_c(4397) = m_p(29/13)^2(1 - 8\alpha)$ with $k = 8 = \dim(\text{adj } SU(3))$, a direct falsification target for LHCb.

The same 29/13 ratio will reappear in Paper 15 as the key GUT scale identity $M_X = \sqrt{M_{R,1}M_{R,3}} = M_{R,2}$ via the $2\sqrt{\alpha}$ neutrino ladder. The cinquefoil is thus *simultaneously* a hadronic topology (observed in pentaquarks) and a GUT topology (breaking $\mathfrak{so}(10)$ via chirality doubling): one knot, two manifestations at two scales.

14 Conclusion

We have presented the Standard Model spectrum as reconstructed within the NWT Casimir framework, using a single continuous input—a mass-scale anchor (m_e , equivalently v_{EW})—plus a discrete topological integer vocabulary. The dimensionless coupling α is derived from the seven-step AB construction (Eq. (1)), not fitted; the master parameter $\kappa_{\text{SM}} \approx 9.844$ is the composite AB-derived quantity, not a free input. Seventy-five particles and seven mixing or coupling parameters match PDG at a median residual of 0.1%; all eighty have been independently recomputed from topological primitives, with every integer derived from the torus-knot crossing algebra (companion Paper 14; Section 10.1). The fine structure constant is derived from first principles via a seven-step Aharonov–Bohm phase calculation, matching CODATA to 7.6 ppm with every factor forced by the trefoil’s D_3 symmetry, BPS topology, or cinquefoil-mode coupling.

The PMNS and CKM CP phases coincide at $\pi - 2$, indicating a shared topological origin. The neutrino hierarchy is a $2\sqrt{\alpha}$ ladder, with right-handed partners cinquefoil phase solitons at 10^{15} – 10^{16} GeV. The pentaquarks $P_c(4312\text{--}4457)$ are cinquefoil hadrons, the first directly-observed visible-sector GUT-topology particles. Their binding multiplet labels the standard NWT topological integer vocabulary.

The numerical simulation program reported in §2.5 has clarified the relationship between the abelian Higgs vortex sector and the Casimir mass formulas. Bogomol’nyi’s non-interaction theorem guarantees that BPS vortex energies are strictly proportional to arc length, with zero crossing energy for any knot topology. The mass hierarchy therefore resides entirely in the gauge-coupling layer (α from the AB-phase derivation) and the Casimir algebra operating on it. The Casimir framework is not an approximation to a “true” vortex-energy calculation—it is the correct and necessary level of description, irreducible to the abelian Higgs energy functional.

Three lines of follow-up work remain. First, testing the predicted additional pentaquark states $P_c(4397)$, $P_c(4499)$, $P_c(4533)$, $P_c(4567)$ in LHCb data ($k = 8, 5, 4, 3$ respectively). Second, the companion Paper 14 derives all fifteen vocabulary integers from torus-knot crossing algebra (crossing counts, braid-group Casimir invariants, toroidal eigenvalues, geodesic norms), demonstrating that the integers are *output* of the knot geometry rather than *input* to the mass formulas. Third, the full GUT

extension and cosmological bootstrap: the cinquefoil is both a visible-sector hadron topology and the $\mathfrak{so}(10)$ -breaking GUT topology. This is the subject of Paper 15, which derives M_{GUT} , the GUT Higgs spectrum, proton decay, monopoles, and leptogenesis from the same κ_{SM} , $\kappa_{\text{GUT}} = 16/3$, and topological integer framework developed here.

The remaining absolute-scale input is itself constrained by a cosmological self-consistency program: Paper 5 [2] derives m_e from the NWT vacuum condensate density, reducing the SM scale-anchor problem from twenty free parameters to a single continuous input plus a constrained vocabulary. Whether the remaining scale input can itself be determined from vacuum cosmology is the open program developed in Paper 15.

A Full Verification Table

Every particle or mixing parameter in the NWT Casimir framework is listed below with its topology assignment, NWT formula, predicted value, PDG (observed) value, fractional residual, and simulation status: $\checkmark$ = independently recomputed from topological primitives (crossing counts, toroidal eigenvalues, Casimir invariants) via the simulation infrastructure of §2.7; **A** = anchor (set by definition). All integers are derived from torus-knot crossing algebra (companion Paper 14). Quantities are in MeV unless otherwise noted (neutrinos in meV, CKM/PMNS elements dimensionless).

Particle	Topology	NWT formula	Predicted	PDG	Resid.	S
Charged leptons						
e	(2, 1, 3, 0)	anchor	0.511	0.511	+0.000%	A
μ	(2, 1, 157, 0)	$m_\tau/(17-25\alpha)$	105.63	105.66	-0.026%	$\checkmark$
τ	(2, 1, 1900, 0)	$25 m_e/(\alpha(1-\alpha)^2)$	1776.5	1776.9	-0.023%	$\checkmark$
Neutrinos (NH; in meV)						
ν_3	cinq +0 Hopf	$(\frac{4}{3}+2\alpha)\alpha^6 v_{\text{EW}}$	50.119	50.15	-0.062%	$\checkmark$
ν_2	cinq +1 Hopf	$2\sqrt{\alpha}(1+\frac{3\alpha}{4})m_{\nu_3}$	8.6097	8.61	-0.004%	$\checkmark$
ν_1	cinq +2 Hopf	$4\alpha(1+\frac{3\alpha}{2})m_{\nu_3}$	1.479	1.48	-0.067%	$\checkmark$
Quarks						
u	(2, 3) light	$225(1+2\alpha)/(54+\alpha) \cdot m_e$	2.1599	2.16	-0.003%	$\checkmark$
d	(2, 3) light	$9(1+2\alpha)m_e$	4.6661	4.67	-0.083%	$\checkmark$
s	(2, 3) strange	$10\alpha^2(1+\alpha)v_{\text{EW}}/\sqrt{2}$	93.391	93.4	-0.010%	$\checkmark$
c	(2, 3) charm	$\alpha v_{\text{EW}}/\sqrt{2}$	1270.5	1270	+0.040%	$\checkmark$
b	(2, 3) bottom	$2\pi\alpha M_Z$	4181	4180	+0.025%	$\checkmark$
t	(2, 3) top	$(1-\alpha)v_{\text{EW}}/\sqrt{2}$	1.728×10^5	1.728×10^5	+0.042%	$\checkmark$
Gauge + Higgs						
γ	Hopf massless	0	0	—	—	A
$g \times 8$	trefoil-bond	0	0	—	—	A
$W^\pm$	Hopf-derived	$M_Z\sqrt{(7-\alpha)/9}$	80378	80379	-0.001%	$\checkmark$
Z^0	Hopf-derived	anchor	91188	91188	+0.000%	A
H^0	Brioschi neu- tral	$(\frac{1}{2}+\alpha+25\alpha^2)v_{\text{EW}}$	1.2523×10^5	1.2525×10^5	-0.012%	$\checkmark$
Light mesons						
π^0	$S^2 \times S^2$	$2m_e/\alpha \cdot (1-5\alpha)$	134.94	134.98	-0.028%	$\checkmark$
$\pi^\pm$	$S^2 \times S^2$	$2m_e/\alpha \cdot (1-\alpha/2)$	139.54	139.57	-0.023%	$\checkmark$
$K^\pm$	$S^2 \times S^2$	$m_{\pi^\pm} \cdot 5/\sqrt{2}$	493.34	493.68	-0.068%	$\checkmark$
η	$S^2 \times S^2$	$10/9 \cdot m_K$	548.16	547.86	+0.054%	$\checkmark$
$\rho(770)$	$S^2 \times S^2$	$11/7 \cdot m_K$	775.25	775.26	-0.001%	$\checkmark$
$\omega(782)$	$S^2 \times S^2$	$27/17 \cdot m_K$	783.54	782.66	+0.113%	$\checkmark$
$K^*(892)$	$S^2 \times S^2$	$29/16 \cdot m_K$	894.18	891.66	+0.283%	$\checkmark$
$\eta'(958)$	$S^2 \times S^2$	$33/17 \cdot m_K$	957.67	957.78	-0.012%	$\checkmark$

Particle	Topology	NWT formula	Predicted	PDG	Resid.	S
$\phi(1020)$	$S^2 \times S^2$	$31/15 \cdot m_K$	1019.6	1019.5	+0.011%	✓
Baryons						
p	S^3 uud	anchor (Paper 6)	938.27	938.27	+0.000%	A
n	S^3 udd	$m_p \cdot (1+2\alpha/9)$	939.79	939.57	+0.024%	✓
Λ	S^3 uds	$19/16 \cdot m_p$	1114.2	1115.7	-0.133%	✓
Σ^0	S^3 uds	$61/48 \cdot m_p$	1192.4	1192.6	-0.021%	✓
$\Delta(1232)$	S^3 quartet	$21/16 \cdot m_p$	1231.5	1232	-0.042%	✓
Ξ^0	S^3 uss	$7/5 \cdot m_p$	1313.6	1314.9	-0.097%	✓
Ξ^-	S^3 dss	$45/32 \cdot m_p$	1319.4	1321.7	-0.171%	✓
Ω^-	S^3 sss	$57/32 \cdot m_p$	1671.3	1672.5	-0.069%	✓
Λ_c^+	S^3 udc	$39/16 \cdot m_p$	2287.0	2286.5	+0.025%	✓
Σ_c	S^3 uuc/ddc	$42/16 \cdot m_p$	2463.0	2454.0	+0.367%	✓
Ξ_c	S^3 usc	$21/8 \cdot m_p$	2463.0	2467.9	-0.202%	✓
Ω_c	S^3 ssc	$23/8 \cdot m_p$	2697.5	2695.2	+0.087%	✓
Λ_b	S^3 udb	$6 \cdot m_p$	5629.6	5619.6	+0.179%	✓
Σ_b	S^3 uub	$99/16 \cdot m_p$	5805.6	5810.6	-0.086%	✓
Ξ_b	S^3 usb	$68/11 \cdot m_p$	5800.2	5797.0	+0.056%	✓
Ω_b	S^3 ssb	$103/16 \cdot m_p$	6040.1	6046.1	-0.099%	✓
Heavy mesons						
D^0	$S^2 \times S^2$ cu	$\sqrt{m_c^2 + (10 m_{\pi^0})^2}$	1853.4	1864.8	-0.613%	✓
$D^\pm$	$S^2 \times S^2$ cd	$\sqrt{m_c^2 + (10 m_{\pi^0})^2}$	1853.4	1869.7	-0.870%	✓
D^*	$S^2 \times S^2$ V	$\sqrt{m_c^2 + (7 m_\tau/8)^2}$	2007.6	2006.8	+0.036%	✓
$D_s^\pm$	$S^2 \times S^2$ cs	$\sqrt{m_c^2 + (11 m_{\pi^0})^2}$	1953.8	1968.3	-0.737%	✓
D_s^*	$S^2 \times S^2$ V	$\sqrt{m_c^2 + (19 m_\tau/20)^2}$	2112.4	2112.2	+0.010%	✓
B^0	$S^2 \times S^2$ bd	$\sqrt{m_b^2 + (24 m_{\pi^0})^2}$	5288.6	5279.7	+0.169%	✓
$B^\pm$	$S^2 \times S^2$ bu	$\sqrt{m_b^2 + (24 m_{\pi^0})^2}$	5288.6	5279.3	+0.175%	✓
B^*	$S^2 \times S^2$ V	$\sqrt{m_b^2 + (13 m_\tau/7)^2}$	5325.9	5324.7	+0.023%	✓
B_s^0	$S^2 \times S^2$ bs	$\sqrt{m_b^2 + (25 m_{\pi^0})^2}$	5372.3	5366.9	+0.101%	✓
B_s^*	$S^2 \times S^2$ V	$\sqrt{m_b^2 + (31 m_\tau/16)^2}$	5415.5	5415.4	+0.002%	✓
$B_c^\pm$	$S^2 \times S^2$ bc	$\sqrt{m_b^2 + (21 m_\tau/8)^2}$	6263.1	6274.5	-0.181%	✓
Charmonium						
$\eta_c(1S)$	$S^2 \times S^2$ cc	$\sqrt{4m_c^2 + (7m_\tau/8)^2}$	2978.7	2983.9	-0.173%	✓
$J/\psi(1S)$	$S^2 \times S^2$ cc	$\sqrt{4m_c^2 + m_\tau^2}$	3100.4	3096.9	+0.113%	✓
χ_{c0}	$S^2 \times S^2$ cc	$\sqrt{4m_c^2 + (9m_\tau/7)^2}$	3416.6	3414.7	+0.057%	✓
χ_{c1}	$S^2 \times S^2$ cc	$\sqrt{4m_c^2 + (15m_\tau/11)^2}$	3510.7	3510.7	+0.001%	✓
$h_c(1P)$	$S^2 \times S^2$ cc	$\sqrt{4m_c^2 + (11m_\tau/8)^2}$	3524.6	3525.4	-0.021%	✓
χ_{c2}	$S^2 \times S^2$ cc	$\sqrt{4m_c^2 + (7m_\tau/5)^2}$	3555.6	3556.2	-0.017%	✓
$\psi(2S)$	$S^2 \times S^2$ cc	$\sqrt{4m_c^2 + (3m_\tau/2)^2}$	3682.0	3686.1	-0.111%	✓
$\psi(3770)$	$S^2 \times S^2$ cc	$\sqrt{4m_c^2 + (11m_\tau/7)^2}$	3774.9	3773.7	+0.031%	✓
Bottomonium						
$\eta_b(1S)$	$S^2 \times S^2$ bb	$\sqrt{4m_b^2 + (17m_\tau/7)^2}$	9409.4	9398.7	+0.114%	✓
$\Upsilon(1S)$	$S^2 \times S^2$ bb	$\sqrt{4m_b^2 + (5m_\tau/2)^2}$	9468.3	9460.3	+0.084%	✓
χ_{b0}	$S^2 \times S^2$ bb	$\sqrt{4m_b^2 + (29m_\tau/10)^2}$	9821.6	9859.4	-0.383%	✓
χ_{b1}	$S^2 \times S^2$ bb	$\sqrt{4m_b^2 + (3m_\tau)^2}$	9916.0	9892.8	+0.234%	✓
χ_{b2}	$S^2 \times S^2$ bb	$\sqrt{4m_b^2 + (3m_\tau)^2}$	9916.0	9912.2	+0.038%	✓
$\Upsilon(2S)$	$S^2 \times S^2$ bb	$\sqrt{4m_b^2 + (28m_\tau/9)^2}$	10023	10023	+0.002%	✓
$\Upsilon(3S)$	$S^2 \times S^2$ bb	$\sqrt{4m_b^2 + (24m_\tau/7)^2}$	10345	10355	-0.098%	✓
$\Upsilon(4S)$	$S^2 \times S^2$ bb	$\sqrt{4m_b^2 + (11m_\tau/3)^2}$	10600	10579	+0.191%	✓
Exotic: pentaquarks (cinquefoil hadrons)						
$P_c(4312)$	$T(2, 5) \ k=10$	$m_p(29/13)^2(1-10\alpha)$	4328.4	4311.9	+0.383%	✓
$P_c(4337)$	$T(2, 5) \ k=9$	$m_p(29/13)^2(1-9\alpha)$	4362.5	4337.0	+0.588%	✓

Particle	Topology	NWT formula	Predicted	PDG	Resid.	S
$P_c(4440)$	$T(2, 5) \ k=7$	$m_p(29/13)^2(1-7\alpha)$	4430.6	4440.3	-0.217%	✓
$P_c(4457)$	$T(2, 5) \ k=6$	$m_p(29/13)^2(1-6\alpha)$	4464.7	4457.3	+0.166%	✓
CKM mixing (dimensionless)						
$\sin^2 \theta_W$		$(2+\alpha)/9$	0.22303	0.22290	+0.060%	✓
V_{us}^2		$7\alpha(1-2\alpha)$	0.05034	0.0503	+0.072%	✓
V_{ub}		$(\alpha/2)(1-\pi\alpha/2)$	0.003607	0.00361	-0.086%	✓
δ_{CP}^{CKM}		$\pi-2 \text{ rad}$	1.1416	1.142	-0.036%	✓
$\alpha_s(M_Z)$		$16\alpha(1+\alpha/3)/(1-\alpha)$	0.1179	0.1179	+0.002%	✓
PMNS mixing (dimensionless)						
$\sin^2 \theta_{13}$		$\alpha(3+\kappa_{SM}\alpha)$	0.02242	0.0224	+0.073%	✓
$\sin^2 \theta_{23}$		$1/2+2\pi\alpha$	0.54585	0.546	-0.027%	✓
$\sin^2 \theta_{12}$		$1/3-(1-\alpha)/(12\pi)$	0.3070	0.307	+0.000%	✓
δ_{CP}^{PMNS}		$\pi-2 \text{ rad}$	1.1416	1.142	-0.036%	✓

Table 4: The complete NWT mass-spectrum and mixing-parameter table. The column S marks each entry as ✓ (simulation-verified: independently recomputed from topological primitives) or A (anchor, set by definition). All 80 non-anchor entries match PDG within 1%, with every integer derived from torus-knot crossing algebra (companion Paper 14).

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Acknowledgements

We thank four AI reviewers—Gemini (Google), Perplexity, ChatGPT (OpenAI), and Claude Mobile (Anthropic, Opus 4.7)—for detailed referee-style critical reviews across three rounds of revision.

Perplexity identified the central “derivation versus fitting” concern and pressed for an explicit separation of derived, constrained, and phenomenological results, leading to the simulation-verification framework of §2.7 and the ✓ markers in Appendix A. ChatGPT, playing the role of a strict referee, flagged the overclaiming of “derives the Standard Model” and pushed for the mechanistic explanation of vocabulary rigidity now in §2.8. Claude Mobile (Opus 4.7) identified the κ_{SM} definitional inconsistency between §2.1 and §2.5 in an earlier draft, proposed the three-scale quark-sector structure in §6, expanded Table 1 from 9 to 15 integers, and caught the $P_c(4670)$ numerical error. Gemini confirmed the Bogomol’nyi non-interaction result and the factual accuracy of the simulation-validation section.

All analysis code, simulation scripts (BPS profile solver, tube-adapted JAX energy functional, multi-knot spectrum scan, crossing-algebra and Jones-polynomial modules), and the L^AT_EX source for this manuscript are available at <https://github.com/JimGalasyn/null-worldtube>.